



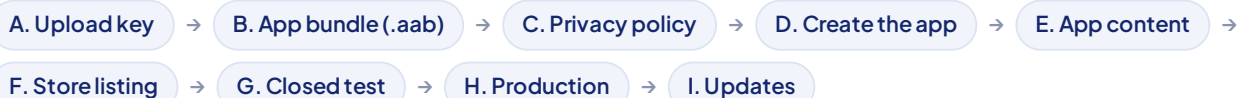
Publishing VCardly on Google Play

The complete A to Z guide, from an empty Play Console account to your app live in the Play Store. Every click, every answer, every text to paste.

App: VCardly: Business Card Scanner · package **com.yasin.vcardly** · version 1.0.0
Prepared 10 October 2026 · Screens and button names in Play Console change from time to time; if a label differs slightly, look for the closest match.

What this guide covers

Follow the parts in order. Each part says how long it takes and ends with what you should see when it is done.



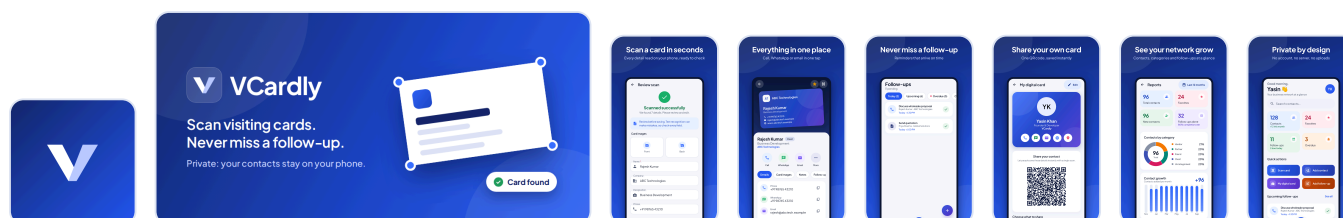
1. **Part A** · Put your upload key into GitHub (5 minutes)
2. **Part B** · Get the Play Store file: the app bundle .aab (10 minutes, automatic)
3. **Part C** · Publish your privacy policy (10 minutes)
4. **Part D** · Create the app in Play Console (5 minutes)
5. **Part E** · App content: the "Set up your app" tasks, with every answer (30 minutes)
6. **Part F** · Store listing: name, descriptions, icon, graphics, screenshots, category, tags (20 minutes)
7. **Part G** · Closed test with 12 testers for 14 days (required for new personal accounts)
8. **Part H** · Production: apply, release, roll out
9. **Part I** · Updating the app later
10. **Part J** · Problems and how to fix them
11. **Part K** · One-page checklist

Words you will meet

Word	What it means
Play Console	Google's website for developers: play.google.com/console . Everything in this guide happens there or on GitHub.
App bundle (.aab)	The file you upload to Google. Google turns it into the right download for each phone. (The .apk files you installed for testing are a different format; Play wants the .aab.)
Upload key	A private digital signature that proves an upload comes from you. You received it as vcardly-upload.jks plus a private text file. Keep both safe.
Play App Signing	Google keeps the final signing key of your app; your upload key only signs what you upload. If you ever lose the upload key, Google support can reset it.
Track	Where a release goes: Internal testing (up to 100 testers, instant), Closed testing (chosen testers), Production (everyone).
versionCode	An internal build number that must be higher in every upload. Our automatic build (CI) raises it for you.
Package name	com.yasin.vcardly . Fixed forever once uploaded; it is already set in the app.

What you already have

Item	Where
Upload key + its 4 values	The two private files sent in our chat: VCardly-upload-key-PRIVATE.txt and vcardly-upload.jks
App icon 512 × 512	store/graphics/icon-512.png
Feature graphic 1024 × 500	store/graphics/feature-graphic.png
6 phone screenshots	store/graphics/screenshots (01 to 06)
All listing texts	Printed in Part F of this guide, ready to copy
Privacy policy page	site/privacy.html (Part C publishes it)



On a computer

Do Parts A to H on a computer (laptop or desktop) with a normal browser. Play Console works on a phone, but uploading files and long forms are much easier on a computer.

Put your upload key into GitHub

The automatic build (CI) signs the Play Store file with your upload key. It reads the key from your repository's encrypted "Secrets", where nobody can see it, not even in the code.

- 1 Open the private file **VCardly-upload-key-PRIVATE.txt** on your computer (any text editor, for example Notepad).

- 2 In the browser, sign in to GitHub and open:

OPEN THIS ADDRESS

<https://github.com/wwds058-spec/vcardly/settings/secrets/actions>

(Or: your repository → **Settings** → left menu **Secrets and variables** → **Actions**.)

- 3 Click **New repository secret**.

- 4 In **Name** type exactly **UPLOAD_KEYSTORE_BASE64**. In **Secret** paste the long line under "1" in the private file (it is one very long line; select all of it). Click **Add secret**.

- 5 Click **New repository secret** again: Name **UPLOAD_KEYSTORE_PASSWORD**, Secret: the password from the private file. **Add secret**.

- 6 Again: Name **UPLOAD_KEY_ALIAS**, Secret: **vcardly-upload**. **Add secret**.

- 7 Again: Name **UPLOAD_KEY_PASSWORD**, Secret: the same password. **Add secret**.

Done when

The Secrets page lists four names: **UPLOAD_KEY_ALIAS**, **UPLOAD_KEY_PASSWORD**, **UPLOAD_KEYSTORE_BASE64**, **UPLOAD_KEYSTORE_PASSWORD**. GitHub never shows the values again; that is normal.

Keep the key private

Never post the private file or the .jks file anywhere public, never put them in the code, never email them to anyone. Save a copy in a password manager or a private drive folder.

Get the Play Store file (.aab)

Now that the key is in place, ask the automatic build to run once more. It makes the signed app bundle and puts it at a fixed download address.

- 1 Open github.com/wwds058-spec/vcardly/actions.
- 2 Click the newest run at the top of the list (named after the latest change).
- 3 Top right: **Re-run jobs** → **Re-run all jobs** → confirm.
- 4 Wait about 10 to 15 minutes until every job shows a green tick ✓ (build, instrumented, release, apk-arm64, ios).
- 5 Download the bundle and its mapping file:

APP BUNDLE (UPLOAD THIS TO PLAY)

<https://github.com/wwds058-spec/vcardly/raw/release-aab/vcardly-release.aab>

MAPPING FILE (UPLOAD LATER, PART G STEP 13)

<https://github.com/wwds058-spec/vcardly/raw/release-aab/mapping.txt>

Done when

You have **vcardly-release.aab** (about 25 to 45 MB) and **mapping.txt** on your computer.

If the release job is red, or the link says 404

A secret name has a typo, or a value was not pasted completely. Check the four names in Part A (they must match exactly, all capitals), fix it with **Update** next to the secret, and re-run. If it still fails, send the red job's name to Claude.

Every later change to the app makes a new bundle at the same address with a higher versionCode, so for updates you just download again.

Publish your privacy policy

Google requires a public web page with your privacy policy. It is already written (it matches exactly what the app does); you fill in three details and switch on free GitHub hosting.

C1. Fill in your details

- 1 Open [the privacy page file on GitHub](#).
- 2 Click the pencil icon  **Edit this file** (top right of the file).
- 3 Press Ctrl+F (Cmd+F on Mac) to find and replace these three placeholders, including the square brackets:

Find	Replace with
[YOUR NAME OR BUSINESS NAME]	your name or your business name, e.g. <i>Yasin Khan</i> or <i>Your Company Pvt Ltd</i>
[YOUR SUPPORT EMAIL]	an email you read, shown publicly (a separate support address is a good idea)
[DATE YOU PUBLISH]	today's date, e.g. <i>10 October 2026</i>

- 4 Click **Commit changes...** → keep "Commit directly to the `claude/happy-goodall-tjw9vc` branch" → **Commit changes**.

C2. Switch on GitHub Pages (free hosting)

- 1 Open github.com/wwds058-spec/vcardly/settings/pages.
- 2 Under **Build and deployment** → Source: **Deploy from a branch**.
- 3 Branch: choose `claude/happy-goodall-tjw9vc`, folder `/ (root)` → **Save**.
- 4 Wait 1 to 3 minutes, then open the page and check your name and email appear:

YOUR PRIVACY POLICY ADDRESS (USED IN PART E)

<https://wwds058-spec.github.io/vcardly/site/privacy.html>

Your repository is public

Anyone can see the app's code on GitHub. That is fine for publishing, and it is what makes free GitHub Pages work. Your key is safe because it is in Secrets, not in the code.

Create the app in Play Console

1 Open play.google.com/console and sign in with your developer account.

2 Left menu **Home** → button **Create app**.

3 Fill in the form:

Field	Answer
App name	VCardly: Business Card Scanner
Default language	English (United States) – en-US
App or game	App
Free or paid	Free
Declarations	Tick Developer Program Policies and US export laws (and Play App Signing terms if shown)

4 Click **Create app**. You land on the app's **Dashboard**.

Free is permanent

Once an app is published as free it can never become a paid download. That is fine for VCardly: Pro will be an optional in-app purchase later, which is allowed in a free app.

Done when

The Dashboard shows sections such as "Start testing now", "Set up your app" and "Release your app", each with a list of tasks. The next parts go through them.

App content: the "Set up your app" tasks

Dashboard → **Set up your app** → **View tasks**. Do each task below; each one ends with **Save**. (All of them are also under left menu **Policy and programmes** → **App content**.)

E1. Privacy policy

Paste the address from Part C → **Save**.

PRIVACY POLICY URL

<https://wvds058-spec.github.io/vcardly/site/privacy.html>

E2. App access

Choose **All functionality in my app is available without any access restrictions**. (There is no login or account in VCardly.) → **Save**.

E3. Ads

Choose **No, my app does not contain ads** → **Save**. This is true for this version: ads are not switched on and the advertising permissions are removed from the app.

E4. Content rating

- 1 **Start questionnaire**. Email address: your support email. Category: **All Other App Types** (VCardly is a business utility) → **Next**.
- 2 Answer **No** to every question about violence, fear, sexuality, nudity, language, controlled substances (drugs, alcohol, tobacco), crude humour and gambling.
- 3 Miscellaneous / interactive questions:

Question (wording may vary)	Answer
Can users interact or exchange content with other users (chat, posting, sharing photos with other users)?	No
Does the app share the user's current physical location with other users?	No
Does the app allow users to purchase digital goods?	No
Does the app contain a web browser / unrestricted internet access?	No
Is the app a web browser or search engine?	No
- 4 **Save** → **Next** → check the summary → **Submit**. Expected rating: suitable for everyone (for example "3+", "Everyone").

E5. Target audience and content

- 1 Target age: tick only **18 and over** → **Next**.
- 2 "Could your store listing unintentionally appeal to children?": **No** → **Next** → **Save**.

E6. Short questions

Task	Answer
News apps	No, my app is not a news app
COVID-19 contact tracing and status apps	My app is not a publicly available COVID-19 contact tracing or status app
Government apps	No
Financial features	My app doesn't provide any financial features
Health apps	My app does not have any health features
AdvertisingID	No (the app does not use advertising ID)

E7. Data safety (the longest form, take it slowly)

"Collected" means **sent off the phone**. VCardly sends no contacts, photos or notes anywhere. The only thing that leaves the phone is anonymous diagnostics from Google's text-recognition library (ML Kit): device model and how long reading a card took, never the image or the text.

1 **Start** (or Manage). Read the overview → **Next**.

2 **Data collection and security:**

Question	Answer
Does your app collect or share any of the required user data types?	Yes
Is all of the user data collected by your app encrypted in transit?	Yes
Which of the following methods of account creation does your app support?	My app does not allow users to create an account
(If asked) Do you provide a way for users to request that their data is deleted?	No (there is no account or server data; everything stays on the phone and is deleted by "Erase all data" or uninstalling)

3 **Data types:** tick only **App info and performance → Diagnostics**. Leave everything else (location, personal info, contacts, photos, financial, messages, device IDs...) unticked → **Next**.

4 **Data usage and handling** → next to *Diagnostics* click **Start**:

Question	Answer
Is this data collected, shared, or both?	Collected (not shared)
Is this data processed ephemerally?	No
Is this data required for your app, or can users choose whether it's collected?	Data collection is required
Why is this user data collected?	Analytics

Save → **Next**.

5 Check the **Preview** (it should say: no data shared; collects App info and performance) → **Save** → **Submit** if asked.

Contacts are not "collected"

The app stores contacts and card photos only on the phone. Do not tick Contacts, Photos, Personal info or Location: Google's form is only about data that leaves the device.

E8. If Play Console asks about permissions

It asks about	Answer
Foreground service	The app does not use foreground services (the permission comes from Android's WorkManager library and declares no service types).
Exact alarms	Used only for follow-up reminders at a time the user chooses.
Camera	Used only to photograph visiting cards while scanning; photos stay on the phone.

Done when

Dashboard → Set up your app shows a green tick next to every task you have done. Store listing (Part F) is the last task in this section.

Store listing: texts, graphics, category, tags

Left menu **Grow users** → **Store presence** → **Main store listing** (sometimes "Grow → Store presence"). Copy each text below into the matching box.

Fl. App details

APP NAME (MAX 30 CHARACTERS)

VCardly: Business Card Scanner

SHORT DESCRIPTION (MAX 80 CHARACTERS)

Scan visiting cards, keep contacts private on your phone, never miss follow-ups

FULL DESCRIPTION (MAX 4000 CHARACTERS)

VCardly turns the visiting cards you collect into organised contacts, and keeps them private on your phone.

SCAN A CARD IN SECONDS

- Point the camera at a card: VCardly finds its edges and straightens it, even when you hold it at an angle.
- Text is read on your phone and sorted into name, job title, company, phones, emails, website and address.
- Check and correct everything before you save. Scan the back of the card too.
- The original card photo is kept with the contact.

STAY ON TOP OF FOLLOW-UPS

- Plan calls, meetings and emails with reminders that arrive on time, even after a restart.
- See what is due today, upcoming and overdue at a glance.

ORGANISE YOUR NETWORK

- Categories, tags and favourites, with fast search and filters.
- Call, WhatsApp or email straight from a contact.
- Reports: new contacts per month, follow-ups by status, categories and tags.

SHARE YOUR OWN CARD

- Create your digital visiting card and share it as a QR code that phones save instantly.
- You choose exactly which details are shared.

PRIVATE BY DESIGN

- No account and no VCardly server: your contacts never leave your phone unless you share or export them.
- Optional App lock with fingerprint, face or PIN, and screen privacy.
- Encrypted backups that you control and keep wherever you choose.

IMPORT AND EXPORT

- Import contacts from .vcf files and export all contacts to one file.
- Export reports as PDF, CSV or Excel.

Accessible: works with TalkBack and the largest font sizes, in light and dark themes.

Free, with no ads and no limits: unlimited scans and every export included.

F2. Graphics (on the same page, scroll down)

Box	Upload this file
App icon	icon-512.png (512 × 512)
Feature graphic	feature-graphic.png (1024 × 500)
Phone screenshots	All six, in order: 01_scan , 02_contact , 03_followups , 04_mycard , 05_reports , 06_home (1080 × 1920)
Video	Leave empty
Tablet / Chromebook screenshots	Leave empty (optional)

Download the images from GitHub: open each link in "What you already have" (page 2) and use the download button (↓) at the top right of the file view.

Click **Save** at the bottom of the page.

F3. Category, tags and contact details

Left menu **Grow users** → **Store presence** → **Store settings**:

Field	Answer
App or game	App
Category	Business
Tags	Manage tags → pick up to 5 from Google's list that fit best. Look for tags such as Business cards , Contacts / Contact management , Scanner / Document scanner , Productivity , CRM . Pick the closest ones Google offers; you cannot type your own.
Email address	your support email (shown publicly)
Phone number	Optional; leave empty
Website	Optional: https://wwds058-spec.github.io/vcardly/site/privacy.html or leave empty
External marketing	Leave on (lets Google show the app in promotions)

Save.

F4. Release notes (used in Part G)

WHAT'S NEW IN THIS RELEASE (MAX 500 CHARACTERS)

First release of VCardly: scan visiting cards with automatic edge detection, keep contacts and follow-up reminders private on your phone, share your own digital card by QR code, and back up with encryption.

Done when

Main store listing and Store settings both show "Saved" with no red errors. The listing becomes public only when a release reaches Production.

Closed test: 12 testers for 14 days

Personal developer accounts created after 13 November 2023 must run a closed test with **at least 12 testers** who stay opted in **for 14 days in a row** before they can publish to everyone. Your Dashboard shows this requirement if it applies to you. (Organisation accounts can skip to Part H.)

G1. Collect 12 or more testers

Ask friends, family or colleagues with an Android phone for the **Gmail / Google account email** they use in the Play Store. Ask for 15 to 20, because some drop out, and the count must stay at 12 or more for 14 days.

G2. Create the closed test

- 1 Left menu **Test and release** → **Testing** → **Closed testing** → on the "Closed testing – Alpha" track click **Manage track**.
- 2 Tab **Countries / regions** → **Add countries / regions** → tick **India** (and any other countries your testers live in) → **Add**.
- 3 Tab **Testers** → **Create email list** → List name **VCardly testers** → paste the emails (comma separated or one per line) → **Save changes**. Make sure the list is ticked.
- 4 Feedback URL or email: your support email → **Save**.
- 5 Still on Testers: find **Join on the web** or **Copy link** (the opt-in link). Copy it and keep it for G4.

G3. Upload the first release

- 1 On the same track click **Create new release**.
- 2 **App integrity / Play App Signing**: if asked, choose **Use Google-generated key** (recommended) → continue. Google keeps the app signing key safe; your upload key only signs uploads.
- 3 **App bundles**: **Upload** → choose **vcardly-release.aab** from Part B. Wait until it shows *1.0.0 (versionCode ...)* with no errors.
- 4 **Release name**: filled in automatically (e.g. *65 (1.0.0)*); keep it.
- 5 **Release notes**: paste the text from F4 between the <en-US> tags that are already there.
- 6 **Next** (or "Save" then "Review release"). Read any warnings. Warnings in yellow are fine; red errors must be fixed (see Part J).
- 7 **Save and publish** / **Start rollout to Closed testing** → confirm.
- 8 Google reviews the first release: this takes from a few hours to about 7 days. You get an email; the status shows **In review**, then **Available to testers**.

G4. Let the testers in

- 1 Once the release is available, send the testers the opt-in link from G2 step 5 with a short message, for example:

MESSAGE TO TESTERS (EXAMPLE)

Hi! Please help me test my new app VCardly. 1) Open this link on your Android phone, signed in with the Google account you gave me: [OPT-IN LINK]. 2) Tap "Become a tester". 3) Tap "Download it on Google Play" and install. Please keep it installed for at least 14 days, open it now and then, and tell me anything that goes wrong. Thank you!

- 2 Watch **Closed testing** → **Testers**: it shows how many have opted in. It must reach 12 or more and stay there.

G5. Upload the mapping file (once per release)

- 1 Left menu **Test and release** → **Latest releases and bundles** → **App bundle explorer** (or "App bundle explorer").
- 2 Select the version you uploaded → tab **Downloads** → **Assets** → **ReTrace mapping file / Deobfuscation file** → **Upload** → choose **mapping.txt** from Part B.

Why

The Play Store version is compressed and its code renamed to make it small. The mapping file lets Play Console show readable crash reports, which Claude can then fix.

Production: release to everyone

1 After 14 days with 12+ testers, Dashboard shows **Apply for production**. Click it.

2 Answer the questions about your test honestly, for example:

Question (wording may vary)	Example answer
How did you recruit testers?	Friends, family and colleagues who collect business cards.
How easy was it to recruit?	Choose the option that matches your experience.
Describe the engagement you received	Testers scanned their own visiting cards, saved contacts, set follow-up reminders and shared QR cards; they reported issues by message, which were fixed in updated test releases.
What changes did you make based on testing?	List real fixes, e.g. "Improved card edge detection on real photos; fixed the camera on some phones."
Who is the intended audience?	Professionals and small-business owners who exchange visiting cards and want to keep contacts organised and private on their phone.
How does the app provide value?	Turns paper visiting cards into contacts in seconds, reminds users to follow up, and keeps everything on the phone without an account.
Expected installs in the first year	Choose a realistic range (e.g. 0 – 10K).

3 **Submit**. Google answers by email, usually within 7 days.

4 When approved: left menu **Test and release** → **Production** → tab **Countries / regions** → add **India** (or **Add all**) → Save.

5 **Create new release** → **Add from library** → pick the bundle already tested (or upload the newest .aab from Part B) → release notes (F4) → **Next**.

6 **Staged rollout**: set **20%** to start (only a part of users get it first, so a problem affects few people). **Save and publish**.

7 After a few days with no crashes (left menu **Monitor and improve** → **Android vitals** → **Crashes and ANRs**), open the release and click **Increase rollout** → **100%**.

Done when

The app's store page is live at play.google.com/store/apps/details?id=com.yasin.vcardly and anyone can install it.

PART I

Updating the app later

- 1 Ask Claude for the change. Every change is pushed to GitHub and the automatic build makes a new bundle with a higher versionCode.
- 2 When all checks are green, download the new **vcardly-release.aab** and **mapping.txt** (same links as Part B).
- 3 Play Console → the track (Closed testing or Production) → **Create new release** → upload → release notes describing what changed → **Save and publish**.
- 4 Upload the new mapping.txt to that version (Part G5).
- 5 For a new user-visible version number (e.g. 1.1.0), ask Claude to change *versionName*.

Switching on ads or VCardly Pro (optional, later)

This first version is completely free with no ads. If you later want income, ask Claude to switch on **AdMob ads** (needs an AdMob account) and/or **VCardly Pro** (create two products in Play Console: *vcardly_pro_lifetime* and *vcardly_pro_yearly*). Then the answers in E3 (Ads), E6 (Advertising ID), E7 (Data safety), the description and the privacy policy must be updated in the same update; Claude prepares all of them.

PART J

Problems and how to fix them

Message or problem	What to do
"Version code ... has already been used"	You uploaded the same bundle twice. Download the newest one (Part B), or ask Claude to trigger a new build.
"Your app must target Android 16 (API level 36)"	Already done (VCardly targets API 36). If it ever appears for a later year, ask Claude to raise it.
"The Android App Bundle was not signed" / "signed with the wrong key"	Check the four GitHub secrets (Part A). Use only bundles from the release-aab link. If you ever lose the key: Play Console → Setup → App signing → Request upload key reset .
Privacy policy URL "invalid" or "not reachable"	Open it in a browser; if it shows 404, re-check GitHub Pages (Part C2) and wait a few minutes.
Advertising ID warning	Answer "No" in E6; this build has no advertising permission (the automatic build checks it every time).
"You need at least 12 testers"	Add more emails to the tester list and make sure each person opened the opt-in link and tapped "Become a tester".
Testers cannot find the app	They must use the same Google account on the phone as the email you added, open the opt-in link first, and their country must be in the track's countries.
Release rejected for "Policy" reasons	Read the email from Google; send the exact text to Claude. Most are fixed by changing a listing text or an App content answer.
Crashes in Android vitals	Make sure mapping.txt is uploaded (G5), then send Claude a screenshot of the crash.

One-page checklist

- ☐ Private key files saved safely (password manager or private drive)
- ☐ 4 GitHub secrets added: UPLOAD_KEYSTORE_BASE64, UPLOAD_KEYSTORE_PASSWORD, UPLOAD_KEY_ALIAS, UPLOAD_KEY_PASSWORD
- ☐ CI re-run, all green; vcardly-release.aab and mapping.txt downloaded
- ☐ privacy.html: name, support email, date filled in
- ☐ GitHub Pages on; privacy policy opens at wwds058-spec.github.io/vcardly/site/privacy.html
- ☐ App created: "VCardly: Business Card Scanner", English (US), App, Free
- ☐ Privacy policy URL saved
- ☐ App access: all functionality available
- ☐ Ads: No
- ☐ Content rating questionnaire submitted (All Other App Types, all No)
- ☐ Target audience: 18 and over; not appealing to children
- ☐ News: No · COVID-19: No · Government: No · Financial: none · Health: none
- ☐ Advertising ID: No
- ☐ Data safety: only App info and performance → Diagnostics, collected, not shared, required, Analytics
- ☐ Store listing: name, short and full description saved
- ☐ Icon, feature graphic and 6 screenshots uploaded
- ☐ Store settings: category Business, tags chosen, support email
- ☐ Closed testing: countries added, tester list with 12+ emails, opt-in link copied
- ☐ First release uploaded (Google-generated app signing key), release notes, rollout started
- ☐ mapping.txt uploaded to the release
- ☐ Testers opted in and installed; 14 days passed with 12+ testers
- ☐ Applied for production; approved
- ☐ Production: countries added, release at 20%, then 100% after a few quiet days

Stuck anywhere?

Take a screenshot of the Play Console page and send it to Claude together with what you clicked. That is all that is needed to work out the next step.